

**ATTN: Chief Information Officer / Office of the Managing Director RE: Request for Correction
Pursuant to the Information Quality Act**

**Marlene H. Dortch, Secretary Federal Communications Commission 45 L Street NE,
Washington, DC 20554**

CC: Chief, Space Bureau RE: ICFS File No. SAT-LOA-20260108-00016

**BEFORE THE
FEDERAL COMMUNICATIONS COMMISSION
WASHINGTON, D.C. 20554**

In the Matter of	)	
	)	ICFS File No.
Space Exploration Holdings, LLC	)	SAT-LOA-20260108-00016
Application for Authorization of	)	
Non-Geostationary Orbit Satellite	)	
System (Orbital Data Center)	)	

**REQUEST FOR CORRECTION PURSUANT
TO THE INFORMATION QUALITY ACT AND
INFORMAL REQUEST FOR COMMISSION ACTION**

William Stewart

April 3, 2026

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1. SUMMARY OF THE FILING

William Stewart ("Petitioner") submits this Request for Correction (RFC) and Informal Request for Commission Action regarding the technical data and scientific models submitted by **Space Exploration Holdings, LLC** ("Applicant") in **ICFS File No. SAT-LOA-20260108-00016**.. This filing demonstrates that the technical exhibits regarding multiple technical issues constitutes "**Influential Scientific Information**" under the 2019 FCC Revised Information Quality Guidelines and OMB Memorandum M-19-15, and is submitted pursuant to the **Information Quality Act (IQA)**.

Because the Commission must rely on this data to formulate its mandatory risk assessments and public interest findings, "it necessarily relies upon this information in evaluating the application, creating the appearance of representing agency views". The Petitioner identifies a systemic failure in the Applicant's technical record, characterized by non-reproducible models and objective gaps in physical reality. Specifically, the Applicant proposes a high-capacity **Orbital Data Center ("ODC")** constellation of one million satellites but fails to account for the mandatory thermal management infrastructure (the "**Radiator Paradox**") required to dissipate heat in a vacuum. By omitting the mass, surface area, and aerodynamic profile of these essential radiators, the Applicant has disseminated "Influential Scientific Information" that is scientifically impossible and fails the IQA standards for **Objectivity, Transparency, and Reproducibility**.

The technical deficiencies detailed in this filing include:

- **The Stefan-Boltzmann Constraint:** The omission of thermal dissipation requirements that fundamentally alter satellite mass and orbital debris profiles.
- **The Debris Factory (0.5% Failure Rate):** Mathematical models that underestimate the cumulative risk of thousands of un-maneuverable satellites.
- **The Twilight Extension Paradox and Cumulative Skyglow:** Non-objective assessments of the constellation's impact on meteorological monitoring and astronomical integrity.
- **The Replacement Flux and Impossible Cadence:** Logistical and physical failures in modeling the continuous launch and re-entry cycle required to maintain the fleet.

Petitioner requests that the Commission:

1. **Halt processing** and compel the Applicant to reconcile these physical and mathematical impossibilities before any authorization is considered.
2. **Require independent technical validation** of the Applicant's "Alpha Phase" data before it is relied upon for commercial authorization.
3. **Provide a formal response within 45 days** as required by the IQA and the Commission's own guidelines.

Failure to address these objective gaps would constitute a violation of the **Administrative Procedure Act (APA)**, as the Commission cannot issue a reasoned decision based on a record containing known physical and mathematical contradictions.

2. STATEMENT OF INTEREST AND PLEADING STATUS

Petitioner is an affected party under the Information Quality Act and a citizen-intervenor in the above-captioned proceeding. As a former Information Technology professional with a background in Systems Analysis and Design, and Telecommunications (Including Satellites), Petitioner's commercial interests and professional obligations are directly impacted by the Commission's reliance on unverified or non-reproducible orbital safety data. The granting of a license for a one-million-satellite constellation based on defective thermal and debris models poses a direct threat to the long-term viability of space-based infrastructure and the safety of the celestial environment, which are essential to the global digital economy.

Petitioner submits this filing as an Informal Request for Commission Action pursuant to 47 C.F.R. § 1.41. Because the flawed information at issue is being relied upon in an active authorization proceeding, the Commission's Information Quality Guidelines command that data quality concerns should be raised directly as comments or filings in the proceeding itself to ensure the administrative record is corrected prior to any final agency action.

3. REQUEST FOR WAIVER OF PROCEDURAL RULES (47 C.F.R. § 1.3)

Petitioner appears before the Commission as an unrepresented (*pro se*) party. To the extent that this filing does not strictly adhere to specialized electronic database formats or pleading specifications enforced upon represented corporate filers, Petitioner respectfully moves for a waiver of such procedural rules pursuant to 47 C.F.R. § 1.3 for "good cause shown."

Good cause exists for such a waiver because Petitioner is a public interest advocate acting *pro se*, and the core purpose of this filing is to put the Commission on notice regarding objective mathematical and physical impossibilities in the record. The Information Quality Act and the Commission's own 2019 Revised Guidelines require that data quality challenges be addressed "within the proceeding itself" to prevent the dissemination of flawed information. Denying the filing of this request on procedural grounds would frustrate the statutory intent of the IQA and risk the integrity of the final agency decision.

4. STATEMENT OF AUTHORITY AND THE IQA MANDATE

This petition invokes a multi-tiered federal mandate:

1. **Primary Statute:** Section 515 of Pub. L. No. 106-554 (44 U.S.C. § 3516 note).
2. **Procedural Mandate:** OMB Guidelines, 67 Fed. Reg. 8452 (Feb. 22, 2002) and **OMB Memorandum M-19-15 (April 24, 2019)**.
3. **Agency Requirement:** 2019 FCC Revised Information Quality Guidelines.

Under these authorities, information that has a clear and substantial impact on important public policies or private sector decisions is categorized as "**Influential Scientific Information.**" For such data, the Commission is required to apply a higher standard of transparency and reproducibility. The IQA triggers the moment the Commission evaluates, references, or relies upon third-party data to support a licensing decision. Reliance on non-reproducible models that omit determinative physical variables (e.g., radiator mass) violates these mandates and renders any resulting order reversible under the Administrative Procedure Act (APA).

5. IDENTIFICATION OF DISSEMINATED INFORMATION SUBJECT TO CORRECTION

Petitioner challenges the following documents, data models, and exhibits submitted by the Applicant:

A. Engineering and Physical Infrastructure Models (The Radiator Paradox)

- **The Stefan-Boltzmann Constraint:** Failure to model the physical dimensions and mass of thermal radiators required for high-capacity orbital computing.
- **The “Baking” Effect:** Inaccurate modeling of localized thermal increases in the 500 km to 2,000 km range resulting from one million active heat-generating nodes.
- **The “3-Kelvin” Deception:** Reliance on deep-space ambient temperature baselines that ignore the operational thermal reality of dense LEO constellations.

B. Orbital Safety and Debris Projections

- **The “Debris Factory” (0.5% Failure Rate):** Non-objective casualty and debris risk assessments that fail to account for the cumulative impact of thousands of unmaneuverable satellites.
- **The “Persistence Gap”:** The omission of data regarding how long fragmentation debris remains in high-altitude orbits (up to 2,000 km) versus lower orbits.
- **The 0.8% Operational Tax:** Failure to disclose the impact of constant atmospheric drag compensation on the long-term structural integrity of the satellite bus.

C. Launch and Replacement Logistics

- **The “Impossible Cadence” and Logistical Failure:** Projections of constellation maintenance that ignore the physical and environmental limits of the global launch infrastructure.
- **The Replacement Flux:** Non-transparent modeling of the continuous "launch-and-burn" cycle required to maintain a million-satellite fleet.

D. Scientific and Environmental Integrity Data

- **The Twilight Extension Paradox and Cumulative Skyglow:** Influential data that underestimates the permanent alteration of the night sky and its impact on astronomical research.
- **Biological Impact:** Omission of reproducible data regarding the effects of large-scale satellite re-entries on upper-atmospheric chemistry.
- **Compounding Risks to Meteorology:** Failure to model the interference and signal degradation to critical disaster-monitoring and weather-tracking frequencies.

E. Developmental Maturity and Prior Art

- **Alpha Phase Production Claims:** Reliance on "Alpha" experimental data as a baseline for full-scale commercial deployment, violating the requirement for sound research methods.
- **Unchallenged Prior Art:** The failure to reconcile the Application with established scientific literature regarding orbital saturation and thermal limits.
- **Internal Energetic Failures:** Empirical evidence (e.g., the fragmentation of Starlink 34343) that contradicts the Applicant’s “0% fragmentation” safety models, rendering the current record inaccurate and subject to immediate correction.

6. MATHEMATICAL AND SCIENTIFIC OBJECTIONS

The Applicant’s technical record fails the IQA’s "Objectivity" and "Reproducibility" standards because it relies on models that omit determinative physical variables. Petitioner provides the following specifications of error:

A. Engineering and Physical Infrastructure Models (The Radiator Paradox)

1. **The Stefan-Boltzmann Constraint:** The Applicant proposes high-density orbital computing without accounting for the Stefan-Boltzmann law ($P = \epsilon \sigma A T^4$), which governs heat radiation in a vacuum. To dissipate the thermal load of an "Orbital Data Center," a satellite requires massive radiator surface area (A). This mass and area are missing from the Applicant's Schedule S, rendering the structural model non-objective.
2. **The "Baking" Effect:** The Applicant fails to provide a reproducible thermal model for the cumulative infrared flux of one million satellites. Concentrating high-wattage compute nodes in the 500 km to 2,000 km range creates a localized "thermal shell" that impacts the structural integrity of neighboring spacecraft—a variable entirely omitted from the record.
3. **The "3-Kelvin" Deception:** The Applicant's safety models appear to rely on the 2.7K background temperature of deep space as a "heat sink" baseline. This is non-objective in Low Earth Orbit (LEO), where solar flux and Earth's albedo significantly limit thermal gradients, making the Applicant's cooling claims physically impossible.

B. Orbital Safety and Debris Projections

1. **The "Debris Factory" (0.5% Failure Rate):** The Applicant claims a high degree of safety but relies on a 0.5% "disposal failure" rate. In a million-satellite constellation, this equates to 5,000 unmaneuverable, "dark" kinetic interceptors. The Applicant's casualty risk data is not "objective" because it treats these failures as isolated incidents rather than a compounding debris-generation system.
2. **The "Persistence Gap":** The Applicant's data fails to distinguish between debris lifespan at 500 km (where atmospheric drag provides remediation) and 2,000 km (where debris persists for centuries). By averaging these risks, the Applicant disseminates misleading information regarding long-term orbital sustainability.
3. **The 0.8% Operational Tax:** Systems analysis dictates that a significant percentage of onboard power must be diverted to station-keeping (drag compensation). The Applicant fails to disclose how this "operational tax" affects the thermal and power budgets of the data center payload, rendering the utility of the data non-transparent.

C. Launch and Replacement Logistics

1. **The "Impossible Cadence" and Logistical Failure:** The Applicant's deployment model requires a launch cadence that exceeds the combined physical capacity of all global launch sites. Information that relies on a physically impossible supply chain lacks "Utility" and "Integrity."
2. **The Replacement Flux:** The Applicant fails to provide a reproducible model for the "Continuous Replacement Cycle." The environmental and safety impact of burning up 200,000+ satellites per year (assuming a 5-year lifespan) is a determinative variable omitted from the atmospheric demise exhibits.

D. Scientific and Environmental Integrity Data

1. **The Twilight Extension Paradox and Cumulative Skyglow:** The Applicant's "DarkSat" mitigation data is non-objective. It fails to account for the cumulative diffuse skyglow of a million objects, which creates a permanent "artificial twilight" that invalidates baseline data for ground-based astronomical research.

2. **Biological Impact:** The record lacks transparent, peer-reviewed data regarding the chemical deposition of alumina and other metallic aerosols in the stratosphere resulting from the unprecedented scale of the proposed re-entry flux.
3. **Compounding Risks to Meteorology:** The Applicant's interference models fail the "Reproducibility" test for weather-tracking frequencies. The signal-to-noise ratio (SNR) projections do not account for the aggregate "side-lobe" emissions of a million-unit constellation.

E. Developmental Maturity and Prior Art

1. **Alpha Phase Production Claims:** The Applicant relies on "Alpha" experimental results to justify a "Production" scale license. Under the IQA, experimental data must be clearly labeled and cannot be used as a "High-Integrity" baseline for permanent commercial authorization.
 2. **Unchallenged Prior Art:** The Applicant ignores established scientific literature regarding "Orbital Saturation" and the "Kessler Syndrome" tipping points. Failure to reconcile the application with existing prior art violates the IQA's requirement for sound research methods.
 3. **Internal Energetic Failures (Starlink 34343):** Recent empirical evidence (e.g., the fragmentation of Starlink 34343) contradicts the Applicant's "0% fragmentation" model. The Applicant's failure to update its safety data in light of these failures renders the current record inaccurate and subject to immediate correction.
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7. REQUEST FOR CORRECTIVE ACTION AND 45-DAY TIMELINE DEMAND

The Information Quality Act regulates the Commission's use of information, not merely its collection. Where the Commission relies on non-government submissions to support a pending authorization, those materials must satisfy standards of objectivity, transparency, and reproducibility. Continued processing of this Application toward authorization constitutes active reliance on the Applicant's technical submissions. Where such submissions omit determinative physical inputs, the Commission cannot lawfully rely on them in reaching a decision.

To the extent such thermal systems are implicitly included, the Application fails to identify or quantify in any attributable or reviewable manner, precluding independent validation. If they are not included, the models do not reflect a physically realizable system. Under either condition, the Commission cannot reliably assess the accuracy or utility of the data upon which the Application depends. See Section 515 of Pub. L. No. 106-554; OMB Guidelines for Ensuring and Maximizing the Quality, Objectivity, Utility, and Integrity of Information Disseminated by Federal Agencies, 67 Fed. Reg. 8452 (Feb. 22, 2002); FCC Information Quality Guidelines (2019).

Because these omissions directly affect satellite mass, surface area, ballistic coefficient, and breakup behavior, they alter the outputs of debris, casualty, and interference analyses in ways that are both material and presently unresolvable on the record. This gives rise to a substantial and material question of fact regarding whether the Application's technical representations correspond to the actual system proposed. See 47 U.S.C. § 309(e).

Where the Commission cannot verify the underlying data or resolve material factual discrepancies, it cannot make the affirmative public interest finding required for authorization under 47 U.S.C. § 309(a). Continued processing on this record would therefore constitute reliance on incomplete and non-reproducible information, contrary to the Commission's obligations under the Information Quality Act and vulnerable to arbitrary and

capricious review under 5 U.S.C. § 706(2)(A). See also *Motor Vehicle Mfrs. Ass'n v. State Farm*, 463 U.S. 29, 43 (1983).

In addition, an application that omits required technical information necessary for evaluation is subject to return as unacceptable for filing. See 47 C.F.R. § 25.112(a); FCC Schedule S Instructions (OMB Control No. 3060-0678).

Accordingly, the Commission lacks a lawful and reviewable basis to proceed toward authorization on the current record.

Where the Commission cannot satisfy the Information Quality Act's standards for influential scientific information, it must withhold reliance on the challenged material, and therefore cannot proceed toward authorization on the current record.

Petitioner therefore moves the Commission to:

1. **Halt further processing of ICFS File No. SAT-LOA-20260108-00016** pursuant to the Commission's obligations under the Information Quality Act and its implementing guidelines, or, in the alternative, return the Application as unacceptable for filing under **47 C.F.R. § 25.112(a)** for failure to include required technical information;
2. **Compel the Applicant to disclose**, in attributable, reviewable, and reproducible form, the full thermal dissipation requirements of the proposed system, including radiator mass, dimensions, materials, and operational characteristics, consistent with the Commission's authority to require complete technical showings under **47 U.S.C. § 308(b) and 47 C.F.R. § 25.114**;
3. **Direct the Applicant to resubmit** all orbital debris, casualty risk, and atmospheric demise analyses using complete and physically representative parameters that incorporate these thermal structures, as required for reasoned technical evaluation and compliance review under Part 25;
4. **Refer the matter for independent technical validation** by Commission engineering staff or appropriate subject-matter experts, consistent with the Commission's obligation to ensure the reliability of the technical record prior to authorization;
5. In the event these deficiencies give rise to unresolved substantial and material questions of fact, designate the Application for further proceedings, including hearing, as required by **47 U.S.C. § 309(e)**. These are determinative factual issues, not policy disagreements, and must be resolved on a complete and verifiable record.

Pursuant to the Commission's Information Quality Guidelines, agencies are expected to respond to requests for correction within a defined timeframe, generally within 45 days. Where, as here, the challenged information is actively relied upon in a pending authorization proceeding, timely resolution is essential to prevent continued reliance on defective or unverified data.

Petitioner therefore requests the Commission provide a formal response within 45 days stating whether it will

- (i) Require correction of the record,
- (ii) Return the Application as unacceptable for filing, or
- (iii) Proceed under a specified procedural mechanism, and the legal and factual basis for that determination.

8. SIGNATURE AND VERIFICATION

I have read the foregoing document and, to the best of my knowledge, information, and belief, there is good ground to support it, and it is not interposed for delay.

Respectfully submitted,

/s/ William Stewart
163 Cuba Hill Rd
Huntington, NY 11743
billbo1970@hotmail.com
Dated: April 3, 2026

9. CERTIFICATE OF SERVICE

I hereby certify that on this 6th day of April, 2026, I caused a true and correct copy of the foregoing **Request for Correction and Informal Request for Commission Action** to be served upon the Applicant's counsel of record via electronic mail and by depositing a copy in the U.S. Mail, First-Class postage prepaid, addressed to:

Via U.S. First-Class Mail & Electronic Email:

William M. Wiltshire, Counsel to SpaceX

wwiltshire@hwglaw.com
Harris, Wiltshire & Grannis LLP
1919 M Street NW, Suite 800
Washington, DC 20036

Via Electronic Mail:

Space Exploration Holdings, LLC

Space Exploration Technologies Corp.
1 Rocket Road
Hawthorne, CA 90250

Attn:

David Goldman, Vice President, Satellite Policy

david.goldman@spacex.com

Jameson Dempsey, Director, Satellite Policy

jameson.dempsey@spacex.com

Cecilia Tenge-Rietberg, Senior Satellite Policy Manager

cecilia.tenge-rietberg@spacex.com